Water: Chemical Tests

Exam code: 0620 & 0971

Download PDF

Written by: Caroline Carroll
Reviewed by: Richard Boole
Updated on 17 January 2026

Chemical tests for water

- The presence of water is commonly tested for using anhydrous cobalt(II) chloride or anhydrous copper(II) sulfate

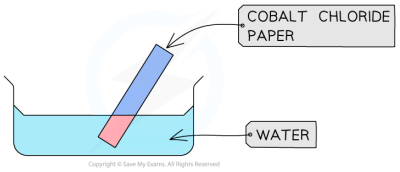
Cobalt(II) chloride

- **Anhydrous** cobalt(II) chloride, $CoCl_2$, is blue
- Hydrated cobalt(II) chloride, $CoCl_2 \bullet 6H_2O$ is pink
- So, anhydrous cobalt(II) chloride can be used to test for water
  - This test is usually done with cobalt chloride paper

anhydrous cobalt(II) chloride + water ⇌ hydrated cobalt(II) chloride

$$CoCl_2\,(s) + 6H_2O\,(l) \rightleftharpoons CoCl_2 \bullet 6H_2O\,(s)$$

- The presence of water causes a colour change from **blue** to **pink**



Cobalt chloride paper changes from blue to pink in the presence of water

Copper(II) sulfate

- **Anhydrous** copper(II) sulfate, $CuSO_4$ is white
- Hydrated copper(II) sulfate, $CuSO_4 \bullet 5H_2O$, is blue
- So, anhydrous copper(II) sulfate can be used to test for water

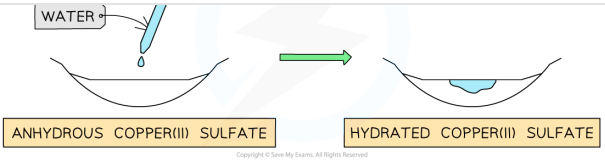
anhydrous copper(II) sulfate + water ⇌ hydrated copper(II) sulfate

$$CuSO_4\,(s) + 5H_2O\,(l) \rightleftharpoons CuSO_4 \bullet 5H_2O\,(s)$$

- The presence of water causes a colour change from **white** to **blue**

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



Anhydrous copper(II) sulfate changes from white to blue in the presence of water

Examiner Tips and Tricks

Core students do **not** need to know the symbol equations.

Purity of water

Testing for purity

- Pure substances boil and melt at specific and sharp temperatures
  - Water has a boiling point of 100 °C and a melting point of 0 °C
- Mixtures have a range of boiling and melting points as they consist of different substances that melt or boil at different temperatures
- Therefore, boiling and melting point data can be used to determine the purity of water
- Impurities tend to **increase the boiling point** of water
  - So, impure water will boil at temperatures above 100 °C
- Impurities tend to **decrease the melting point** of water
  - So, impure water will melt at temperatures **below 0 °C**

What is distilled water?

- Distilled water is used in practical chemistry rather than tap water because it contains fewer chemical impurities
- Tap water contains dissolved ions and other impurities
  - These impurities could interfere with chemical reactions and affect results
- Distilled water is made by heating water to form vapour and then condensing it back to a liquid
  - This removes most dissolved impurities

Teacher only

New! Exclusive Teacher-Only Questions

Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Flashcards

Was this revision note helpful? Yes No

Previous

Next

Course

Study tools

All topics

Resources

Learning Hub

Scholarship

Join

Past Papers

Solution Banks

Sitemap

Dashboard

Overview

Brand ambassador

Profile

Subscription

Settings

Log out

Company

About us

Exam Specificity

Content Quality

Promotions

Jobs

Terms

Privacy

Cookie Policy

Help and Support

Subjects

Biology

Chemistry

Physics

Maths

Geography

English Literature

Psychology

All Subjects